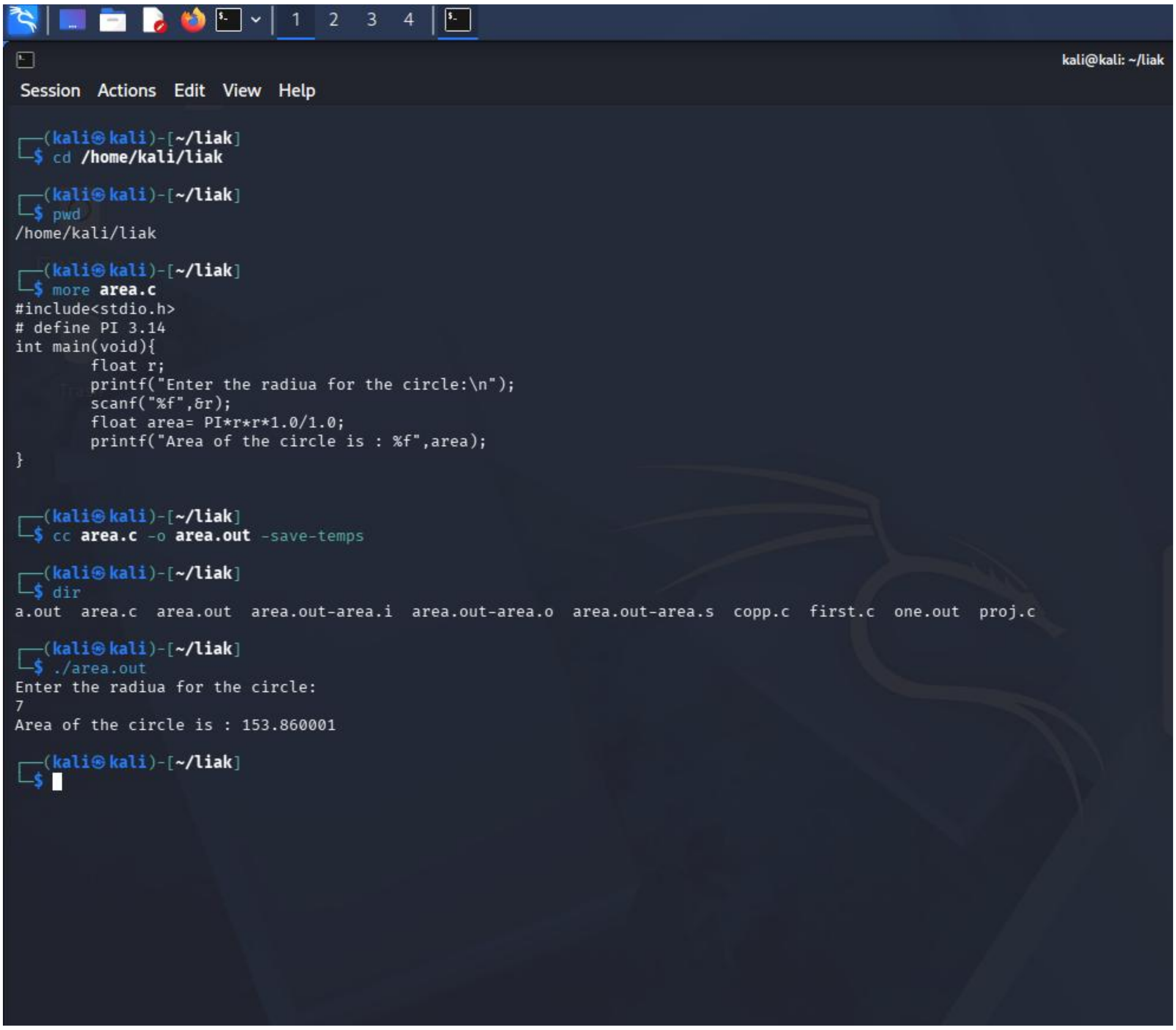
Experiment-2

Q. Finding area of a circle after getting input for the radius from the user.

A.



When we add the command -save-temps in the terminal we get all the temporary stages that the .c file goes through before getting converted to the final linked executable file in the machine. As we can see when we put the -save-temps file we get .i, .s and the .o file which stand for the following:

1. .i file: This file format stand for the preprocessed file format in which the computer removes all the comment that the user has put in, include all the # and import the actual numbers and define the macros. In the end, It pastes the <stdio.h> file's actual code before the actual written code so that the compiler understands the thing. Also, this file is human readable, as no major change for computer understanding took place

2. .s file: This file is in the assembly code format, this file helps the code to communicate with the CPU. Computer took the .i file and translated it to the computer readable format into the assemble language. It contains the CPU-memory related translations.
3. .o file: This file format contains the code in raw binary code for the computer to understand. It compiles all the mathematics and the logic in the code. But you can't run this file as this has not been converted into the linked executable file that the system requires to finally show the output to the user on his screen.
4. .out/.exe file: This is the final step, this is the linked executable file for the compiler that shows the output of the code. .o is the machine code, user can't understand it efficiently. The compiler builds the parts, the linker assembles them into the .out file, and the .out file gives the operating system everything it needs to actually run your software.

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